Universidad Nacional Autónoma de México
Facultad de Ingeniería

Laboratorio de Diseño Digital VLSI

Proyecto final: sintetizador musical

Primer avance

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Tarjeta EP2C5T144 (Cyclone II EP2C5T144C8)
Quartus II 13.0 SP1 Web Edition

Repositorio: `https://github.com/samisal123/robot-seguidor-luz-vlsi`

10 de noviembre de 2026

1. Diagrama general de bloques

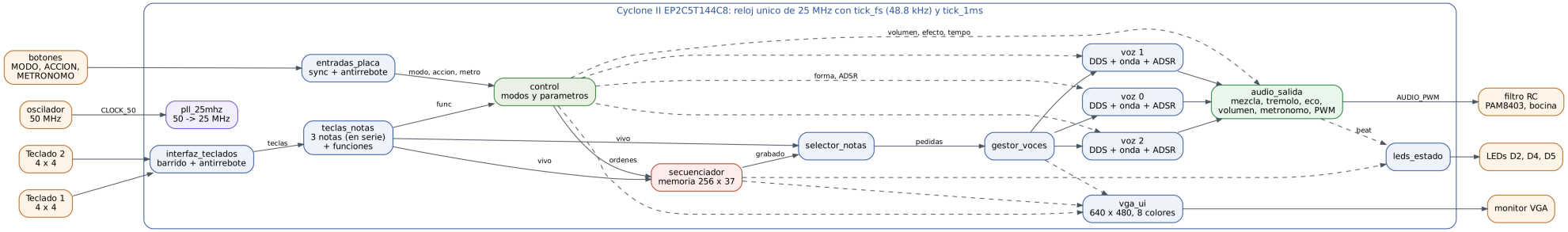


Figura 1: Diagrama general de bloques del sintetizador en la Cyclone II.

2. Proyecto en Quartus

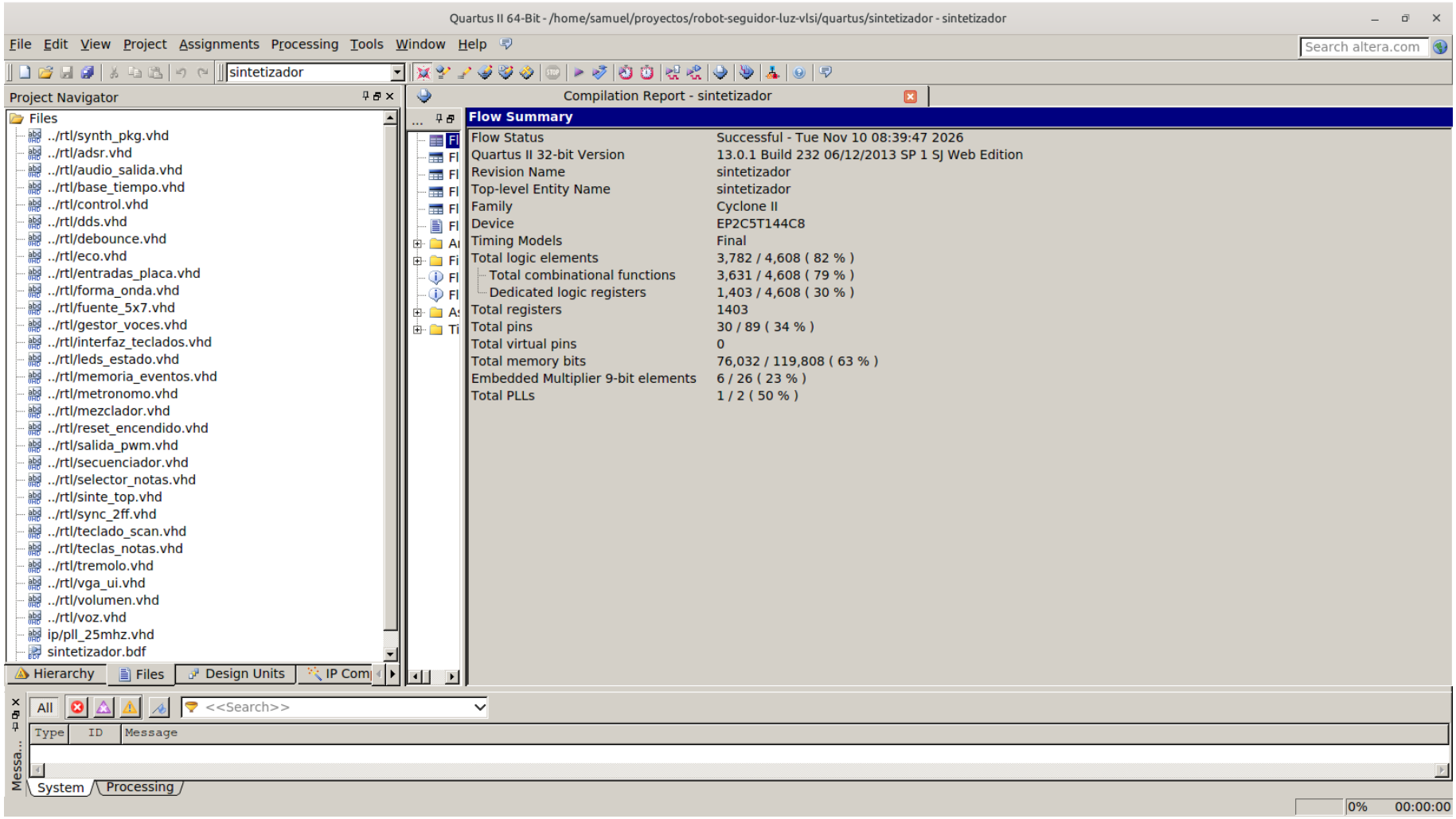


Figura 2: Proyecto en Quartus II 13.0 SP1 para la EP2C5T144C8 (pestaña Files).

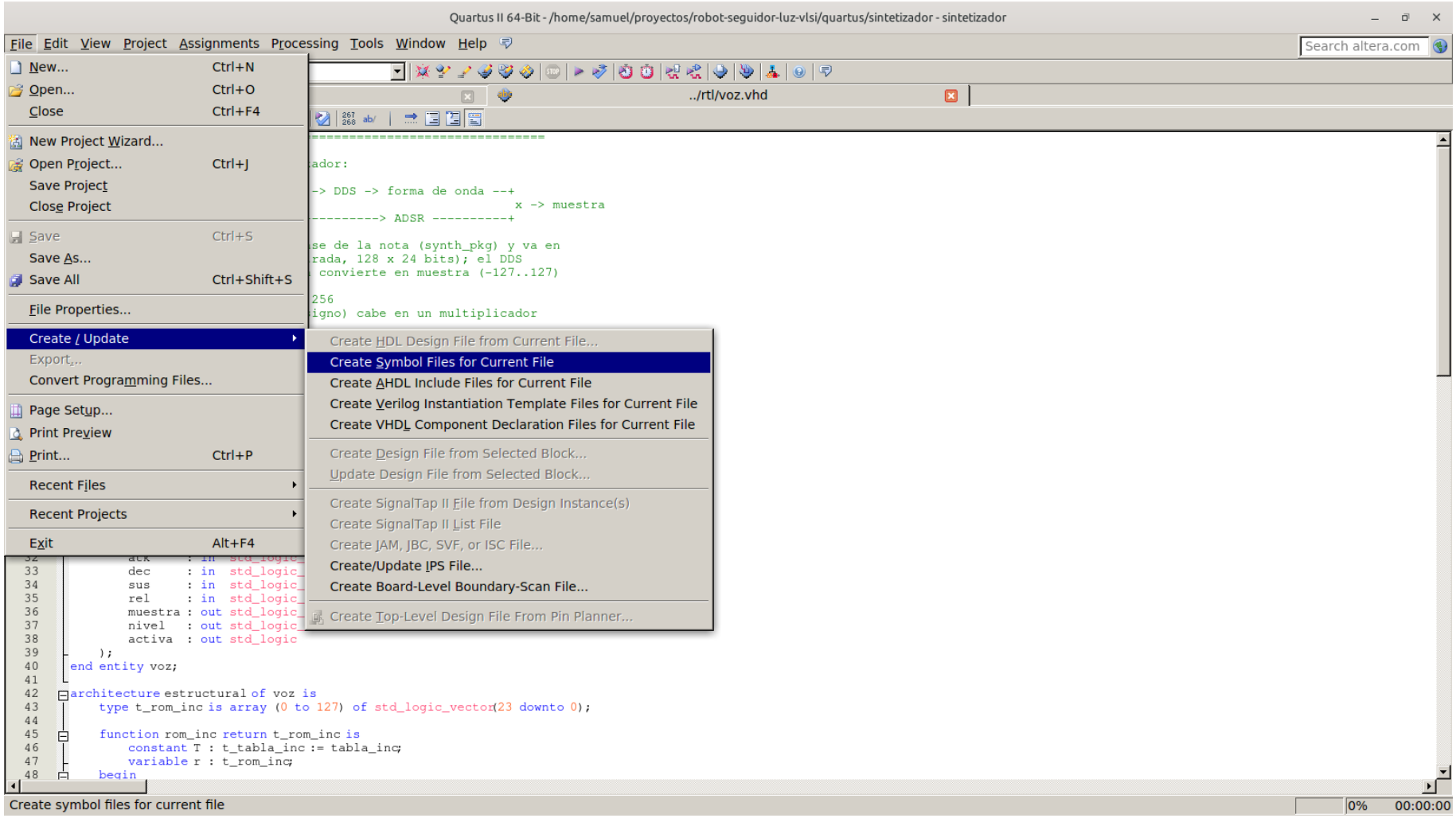


Figura 3: Creación del símbolo a partir del archivo .vhd (File, Create / Update, Create Symbol Files for Current File).

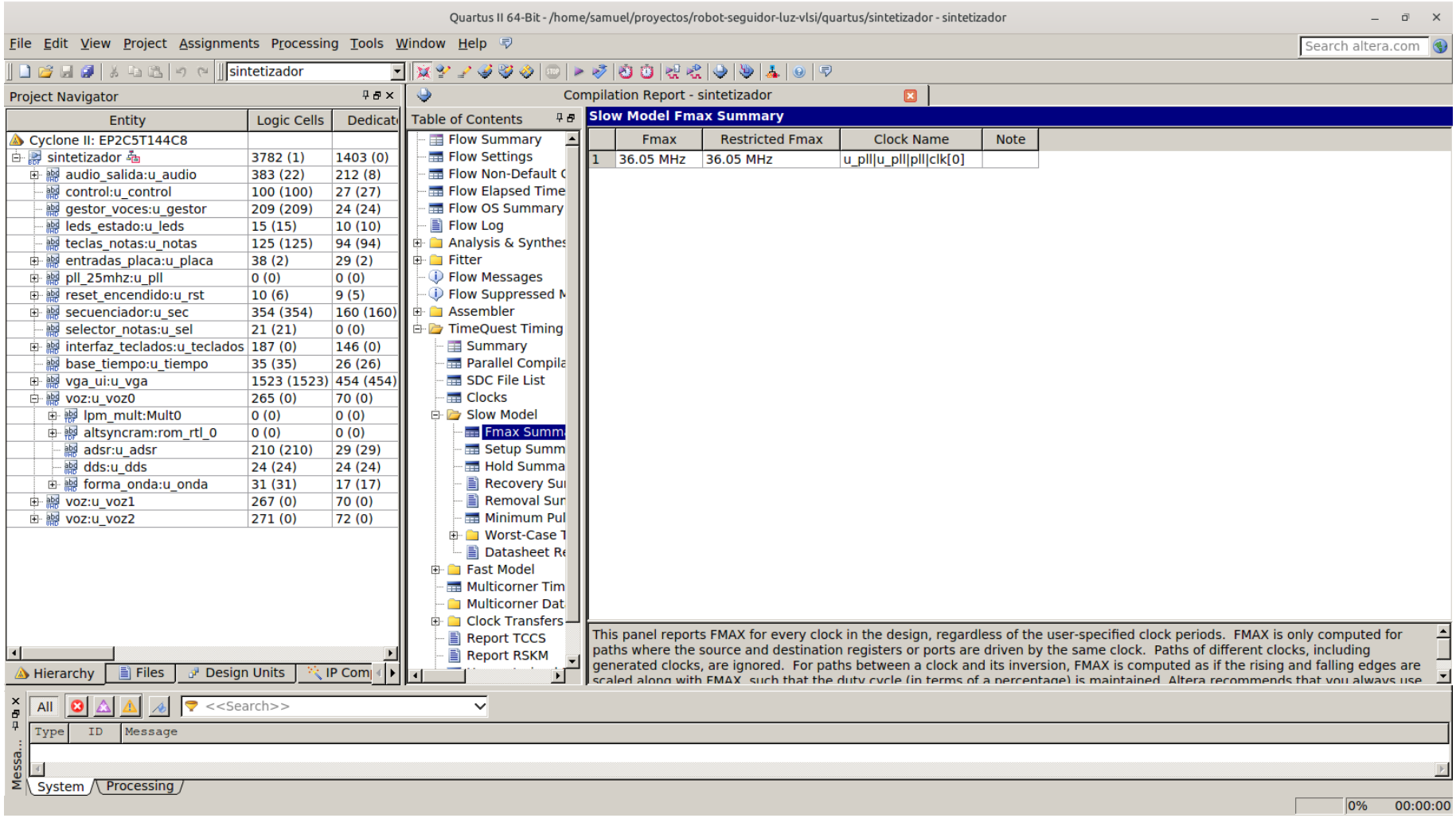


Figura 4: Jerarquía del proyecto con el diagrama de bloques `sintetizador` como nivel superior.

3. Diagrama de bloques en Quartus

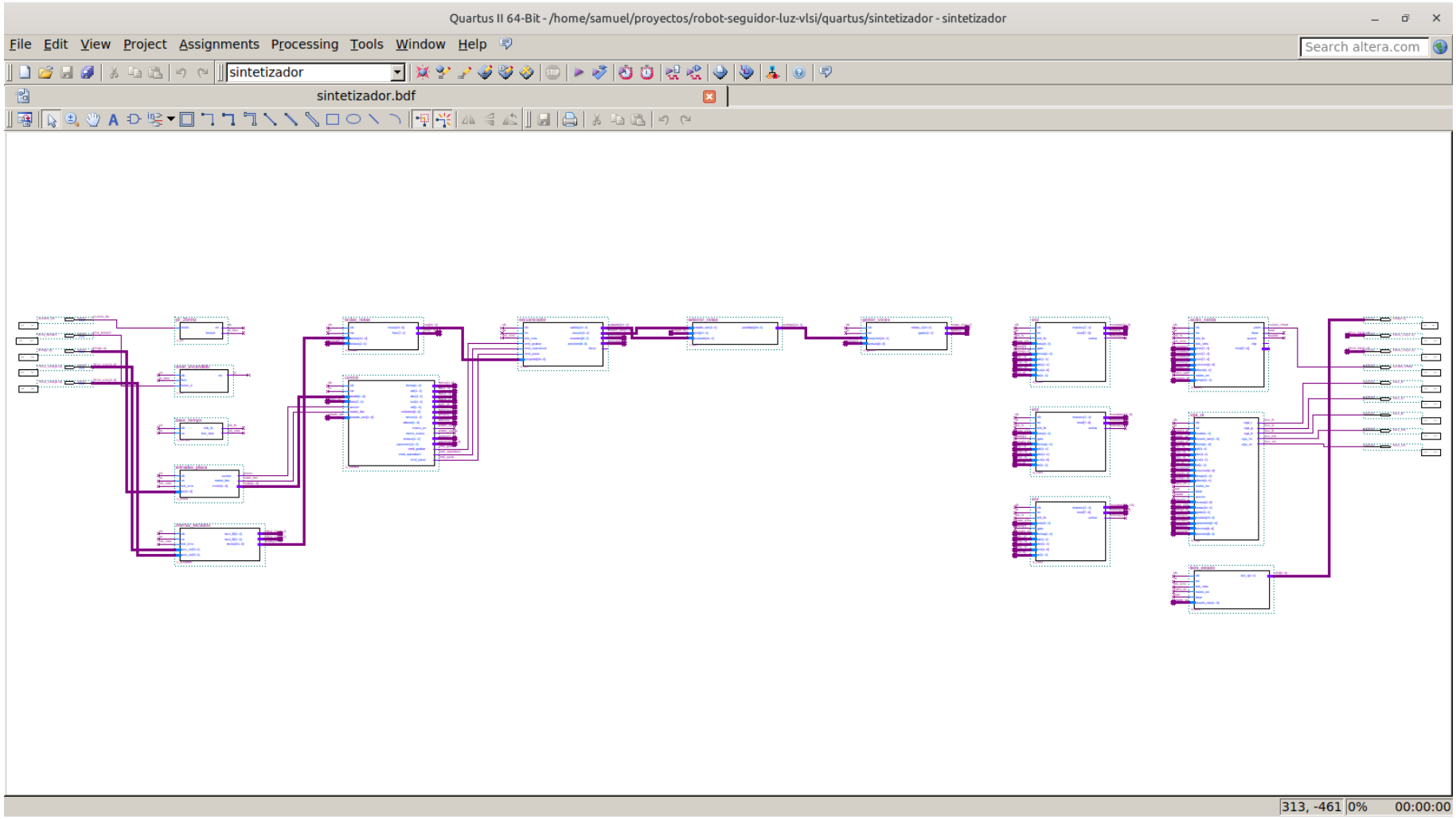


Figura 5: Diagrama de bloques del sintetizador (`sintetizador.bdf`, nivel superior).

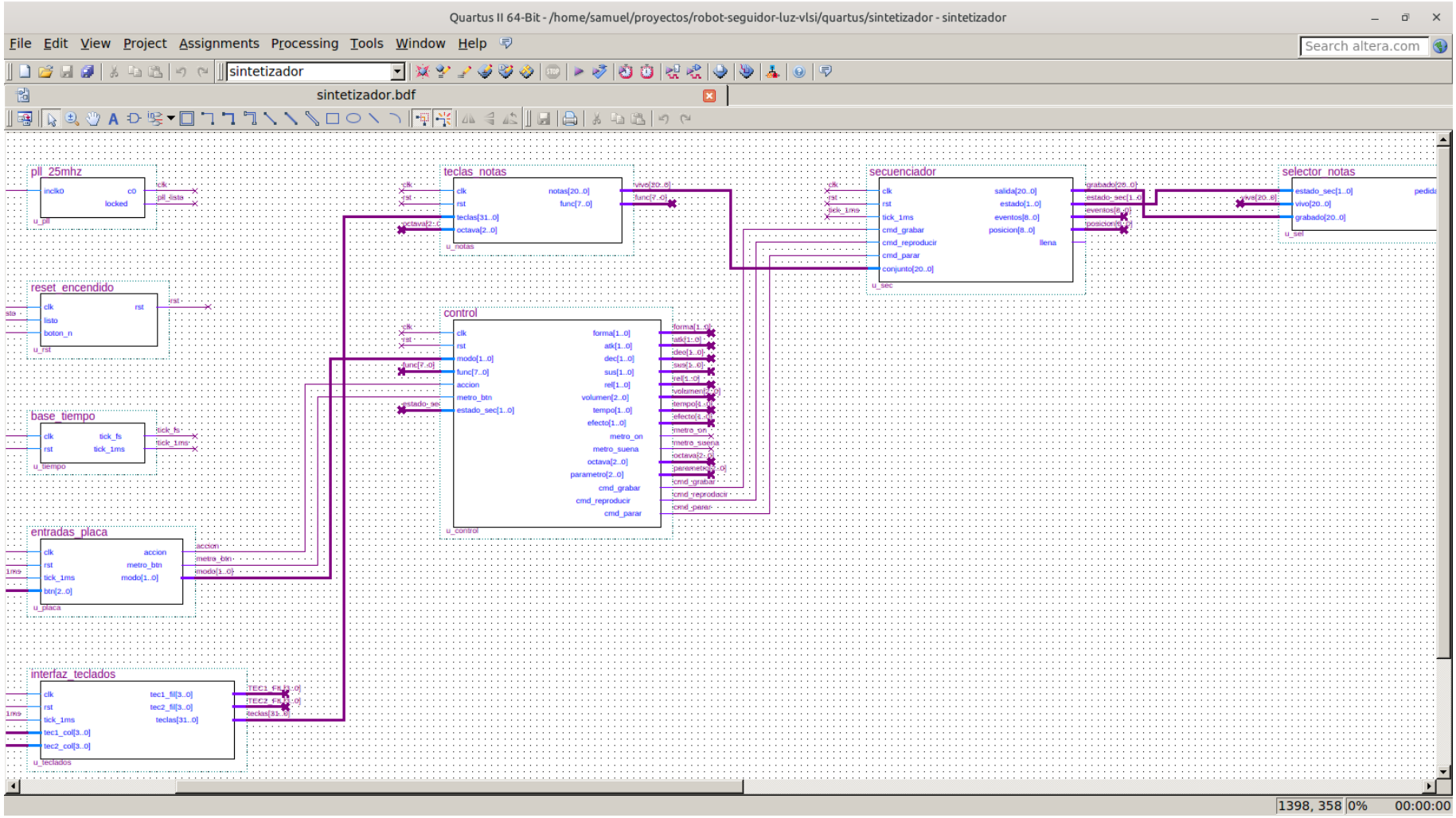


Figura 6: Mitad izquierda: reloj, reset, botones, teclados, control, secuenciador y selector de notas.

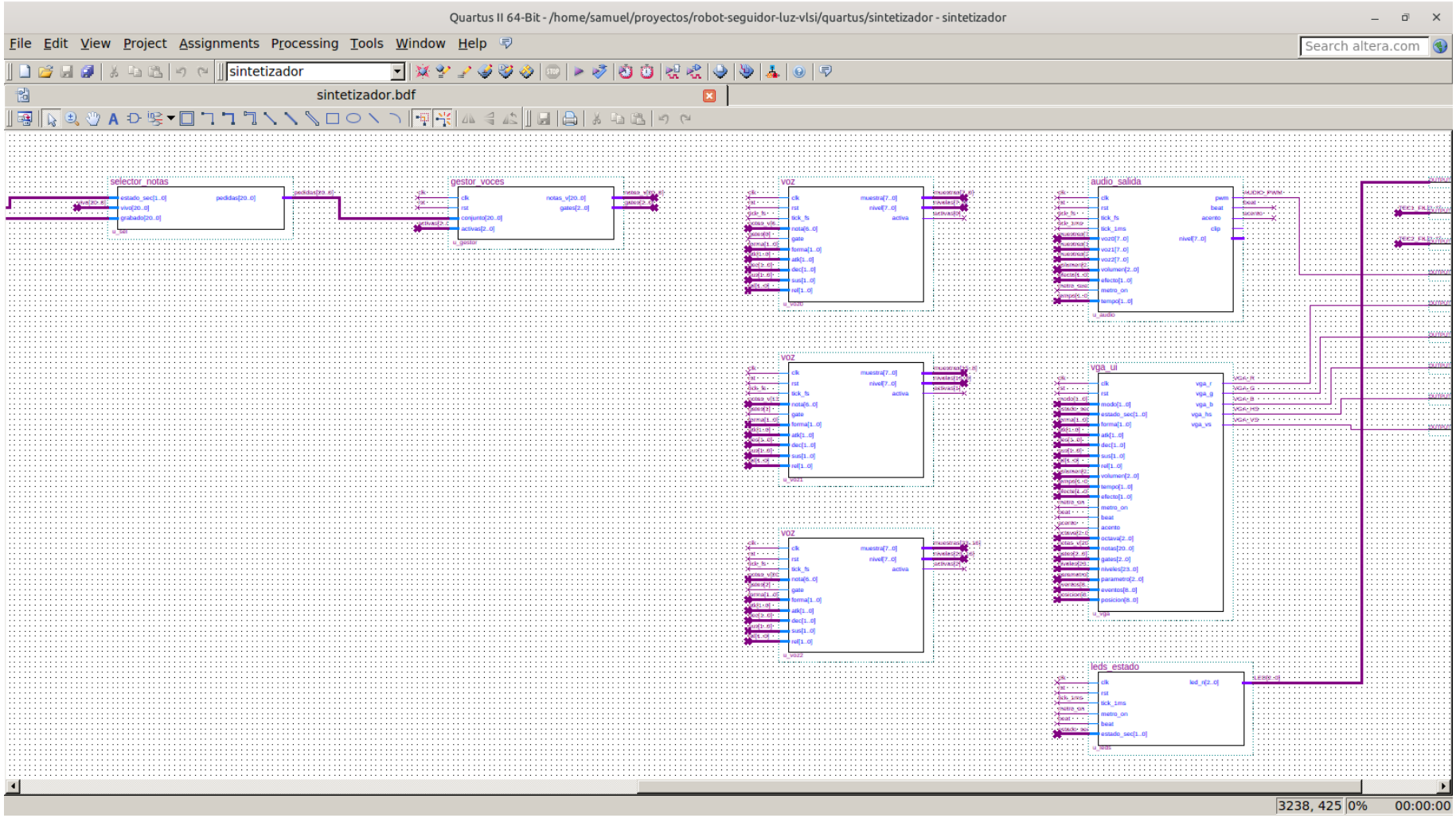


Figura 7: Mitad derecha: gestor de voces, las 3 voces, salida de audio, VGA y LEDs.

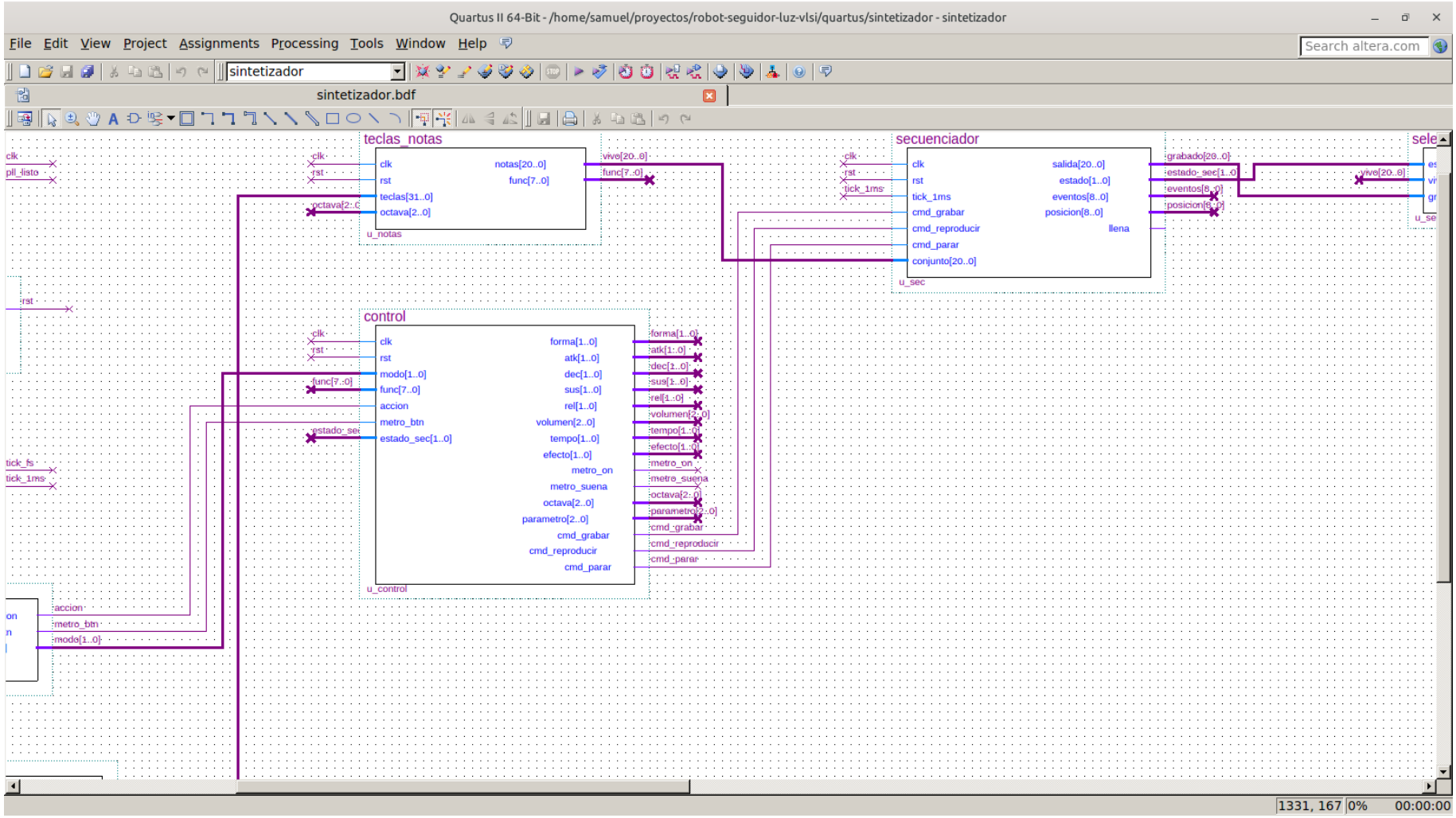
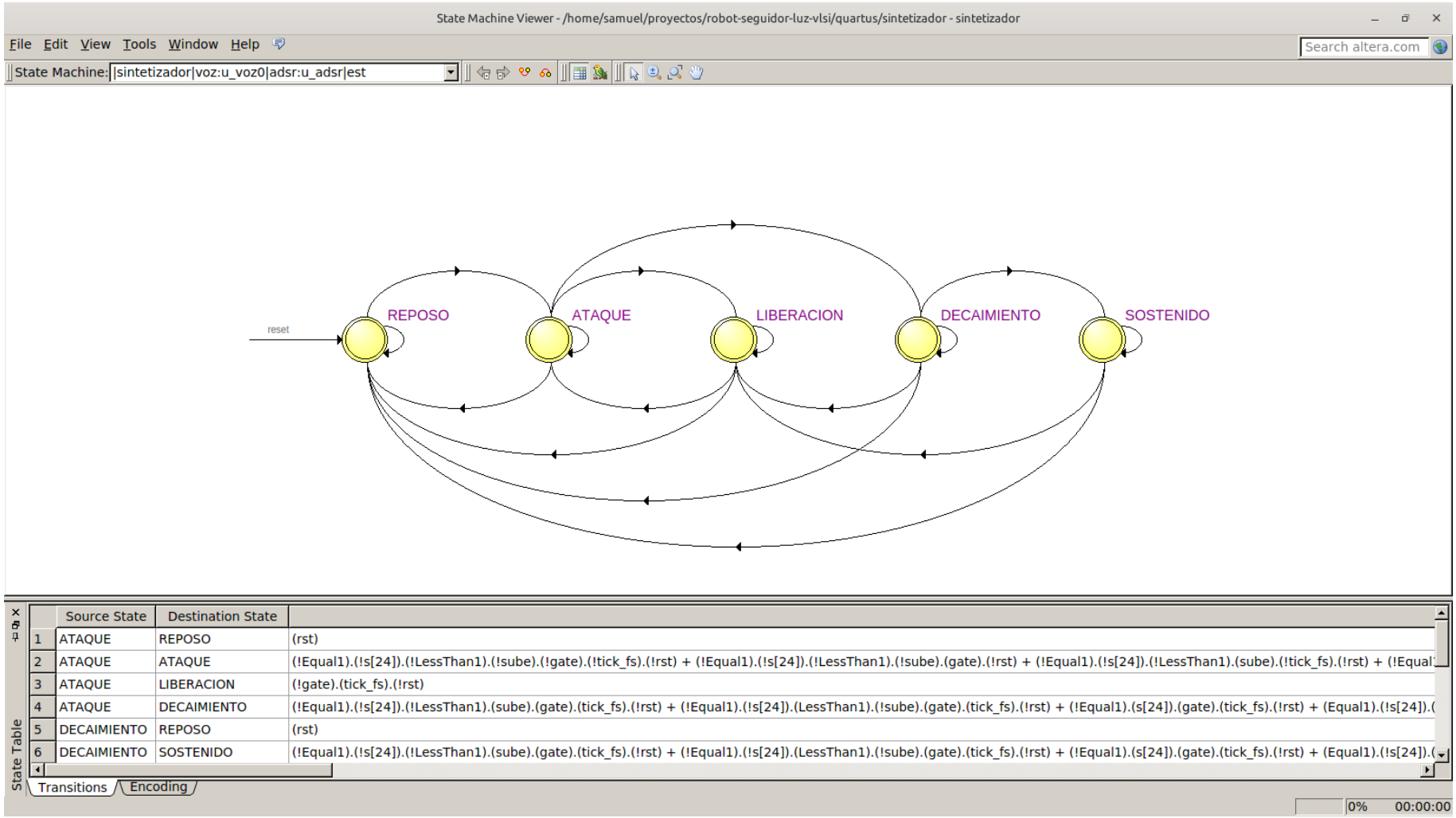


Figura 8: Símbolos de los módulos con sus puertos.

4. Máquina de estados



	Source State	Destination State	
1	ATAQUE	REPOSO	(rst)
2	ATAQUE	ATAQUE	(!Equal1).(!s[24]).(!LessThan1).(!sube).(!gate).(!tick_fs).(!rst) + (!Equal1).(!s[24]).(!LessThan1).(!sube).(gate).(!rst) + (!Equal1).(!s[24]).(!LessThan1).(sube).(!tick_fs).(!rst) + (!Equal
3	ATAQUE	LIBERACION	(!gate).(tick_fs).(!rst)
4	ATAQUE	DECAIMIENTO	(!Equal1).(!s[24]).(!LessThan1).(sube).(gate).(tick_fs).(!rst) + (!Equal1).(!s[24]).(LessThan1).(!sube).(gate).(tick_fs).(!rst) + (!Equal1).(s[24]).(gate).(tick_fs).(!rst) + (Equal1).(!s[24]).(
5	DECAIMIENTO	REPOSO	(rst)
6	DECAIMIENTO	SOSTENIDO	(!Equal1).(!s[24]).(!LessThan1).(sube).(gate).(tick_fs).(!rst) + (!Equal1).(!s[24]).(LessThan1).(!sube).(gate).(tick_fs).(!rst) + (!Equal1).(s[24]).(gate).(tick_fs).(!rst) + (Equal1).(!s[24]).(

Figura 9: Máquina de estados del ADSR de cada voz (State Machine Viewer).

5. Compilación

Cuadro 1: Resumen de la compilación (Flow Summary de Quartus y TimeQuest, modelo lento).

Concepto	Valor
Dispositivo	EP2C5T144C8 (Cyclone II)
Elementos lógicos	3 782 de 4 608 (82 %)
Funciones combinacionales	3 631 de 4 608 (79 %)
Registros	1 403 de 4 608 (30 %)
Bits de memoria	76 032 de 119 808 (63 %)
Multiplicadores de 9 bits	6 de 26 (23 %)
PLL	1 de 2
Pines	30 de 89 (34 %)
Reloj del diseño	25 MHz (PLL a partir de 50 MHz)
Fmax	36.05 MHz
Holgura de setup / hold	12.261 ns / 0.499 ns

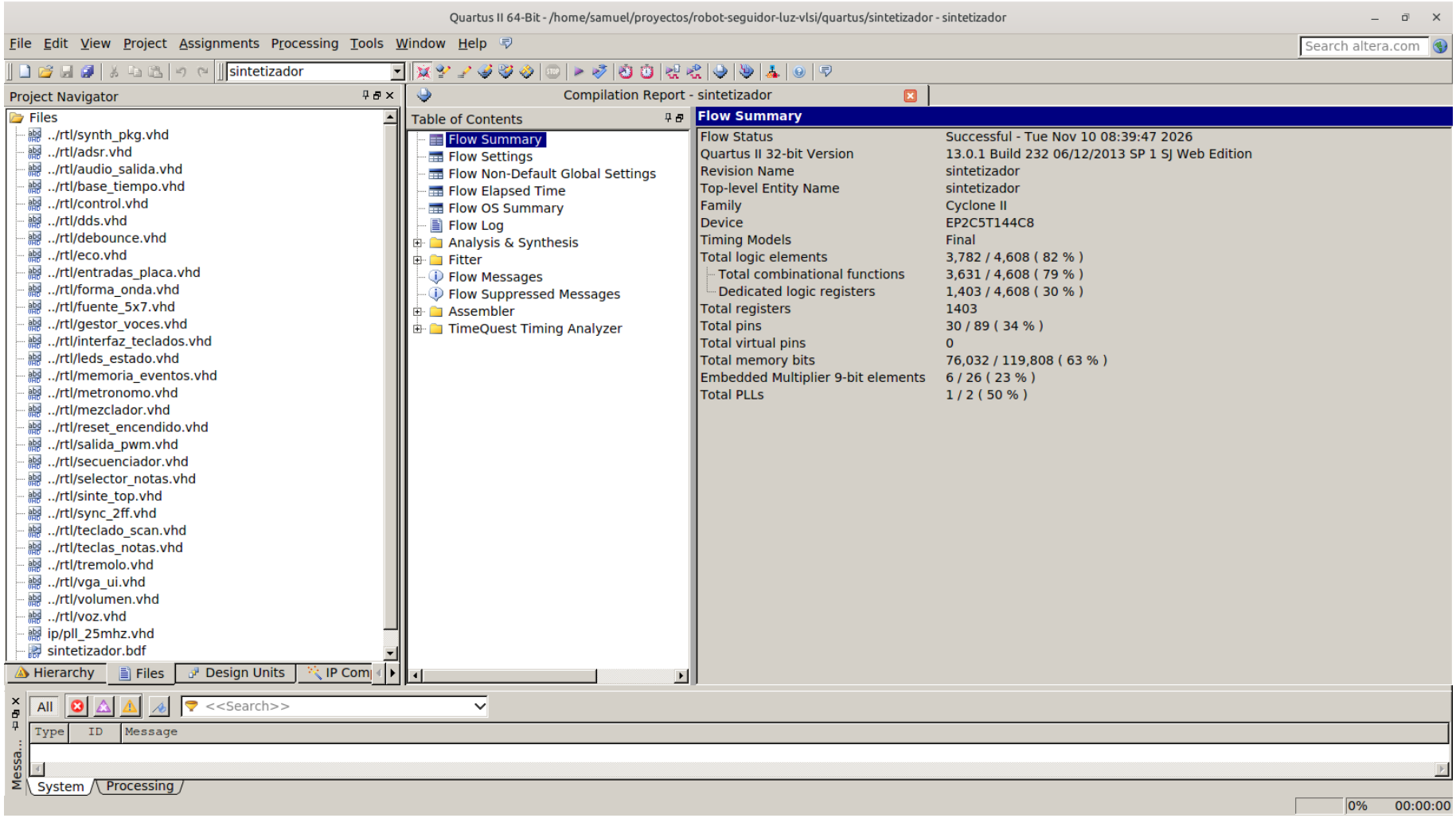


Figura 10: Resumen de la compilación (Flow Summary).

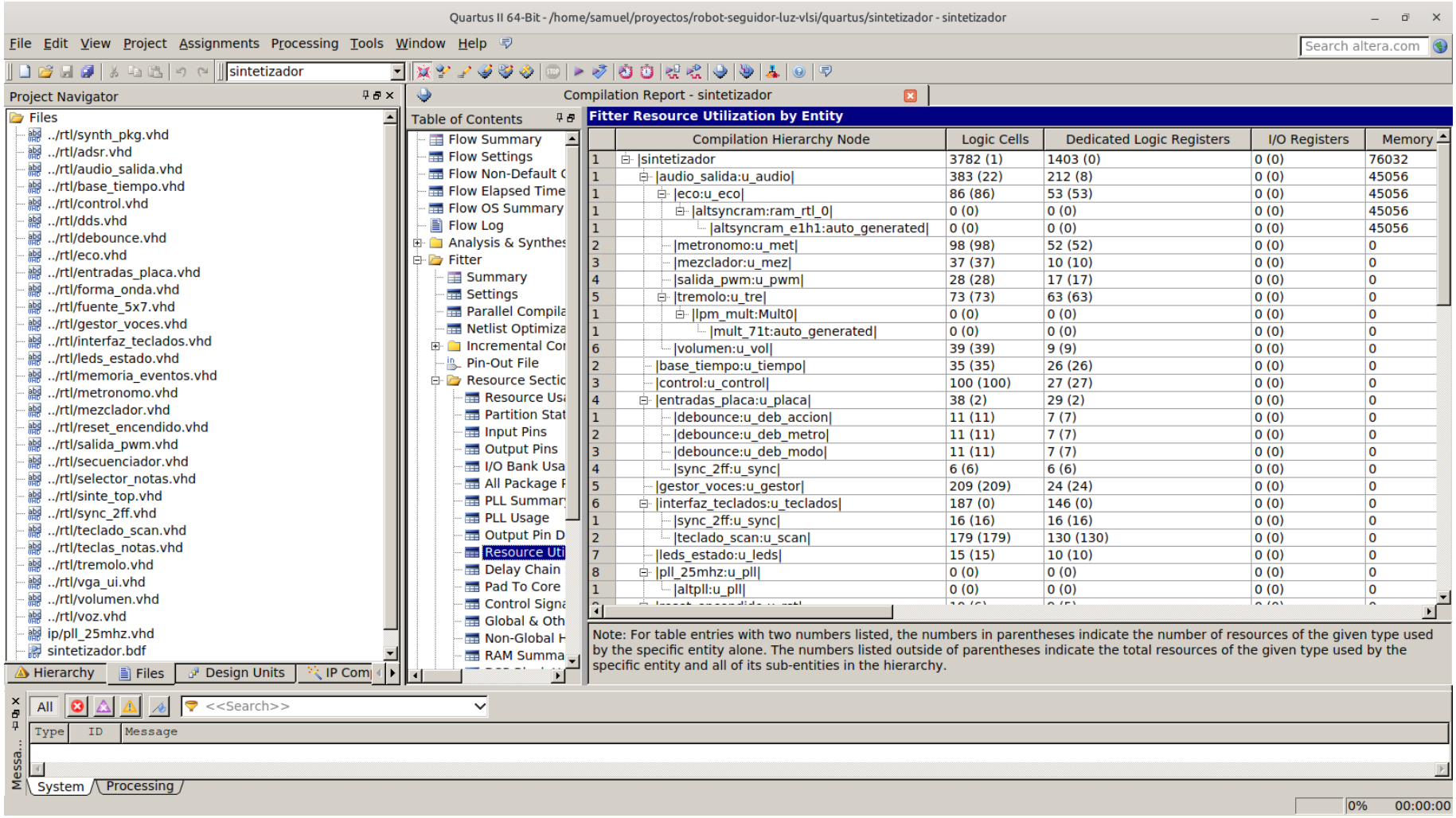


Figura 11: Recursos por módulo (Fitter, Resource Utilization by Entity).

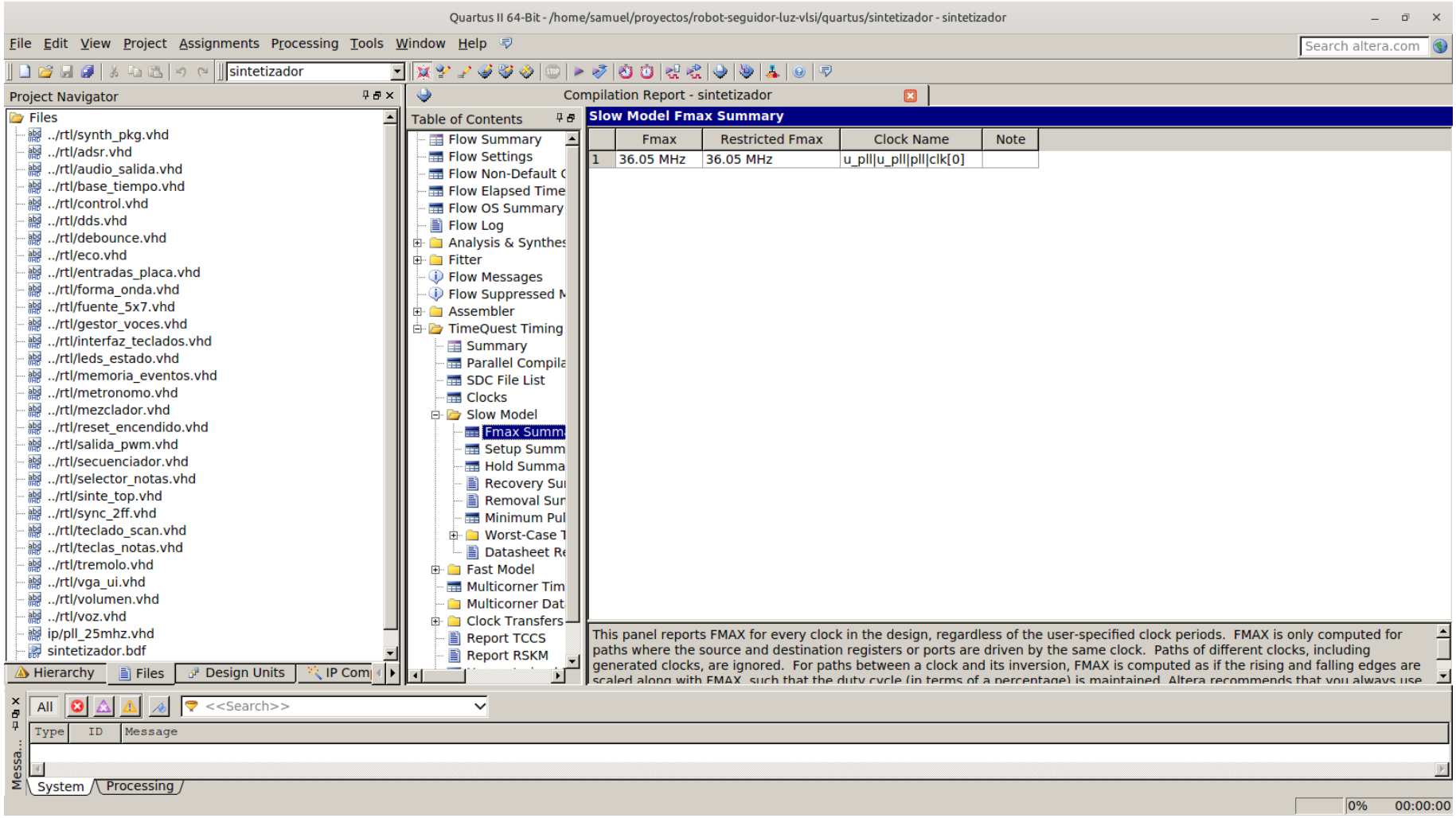


Figura 12: Frecuencia máxima (TimeQuest, Slow Model, Fmax Summary).

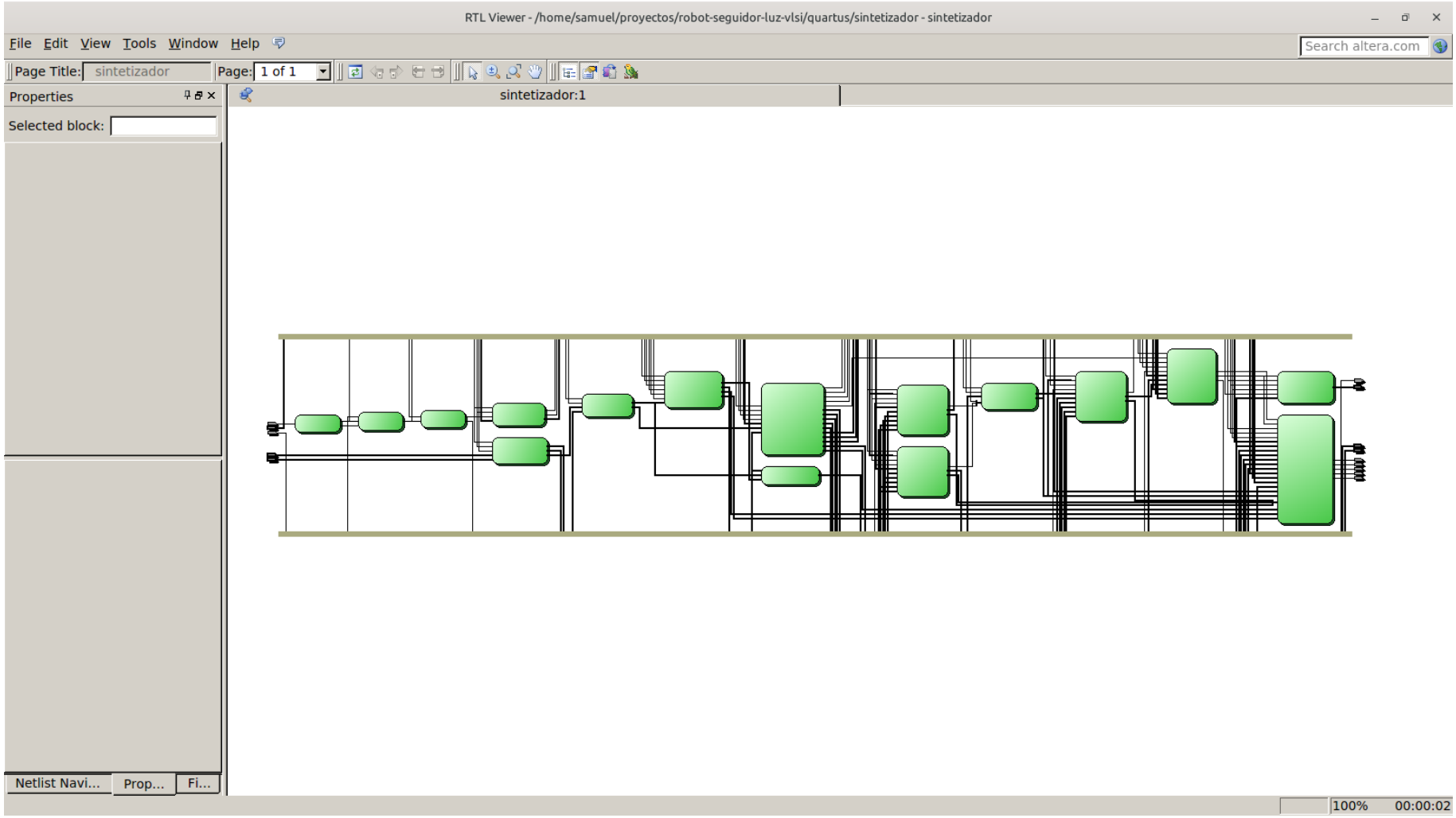
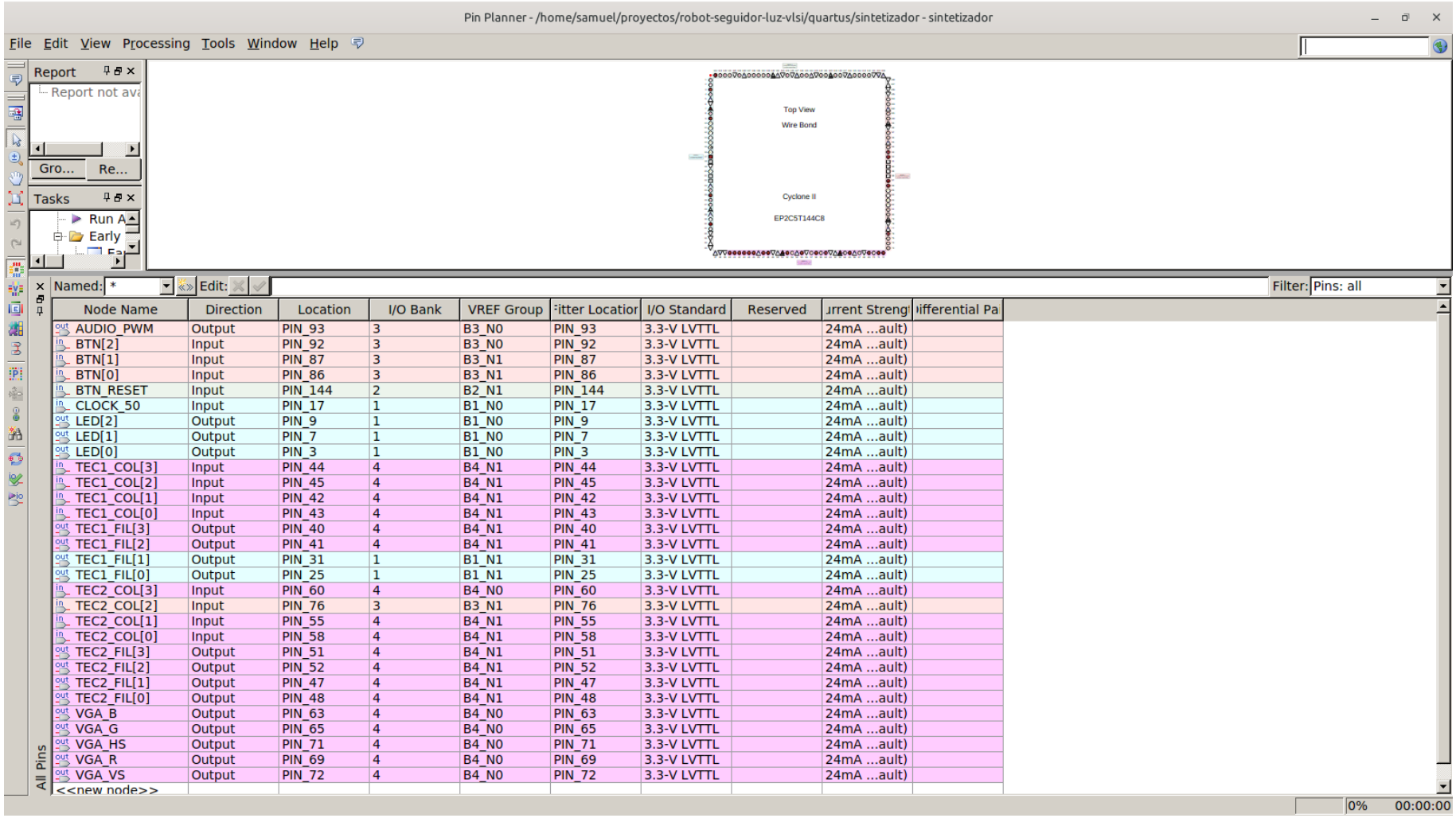


Figura 13: Vista RTL del nivel superior.



Node Name	Direction	Location	I/O Bank	VREF Group	itter Locatior	I/O Standard	Reserved	urrent Streng	ifferential Pa
AUDIO_PWM	Output	PIN_93	3	B3_N0	PIN_93	3.3-V LVTTL		24mA ...ault)	
BTN[2]	Input	PIN_92	3	B3_N0	PIN_92	3.3-V LVTTL		24mA ...ault)	
BTN[1]	Input	PIN_87	3	B3_N1	PIN_87	3.3-V LVTTL		24mA ...ault)	
BTN[0]	Input	PIN_86	3	B3_N1	PIN_86	3.3-V LVTTL		24mA ...ault)	
BTN_RESET	Input	PIN_144	2	B2_N1	PIN_144	3.3-V LVTTL		24mA ...ault)	
CLOCK_50	Input	PIN_17	1	B1_N0	PIN_17	3.3-V LVTTL		24mA ...ault)	
LED[2]	Output	PIN_9	1	B1_N0	PIN_9	3.3-V LVTTL		24mA ...ault)	
LED[1]	Output	PIN_7	1	B1_N0	PIN_7	3.3-V LVTTL		24mA ...ault)	
LED[0]	Output	PIN_3	1	B1_N0	PIN_3	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[3]	Input	PIN_44	4	B4_N1	PIN_44	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[2]	Input	PIN_45	4	B4_N1	PIN_45	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[1]	Input	PIN_42	4	B4_N1	PIN_42	3.3-V LVTTL		24mA ...ault)	
TEC1_COL[0]	Input	PIN_43	4	B4_N1	PIN_43	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[3]	Output	PIN_40	4	B4_N1	PIN_40	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[2]	Output	PIN_41	4	B4_N1	PIN_41	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[1]	Output	PIN_31	1	B1_N1	PIN_31	3.3-V LVTTL		24mA ...ault)	
TEC1_FIL[0]	Output	PIN_25	1	B1_N1	PIN_25	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[3]	Input	PIN_60	4	B4_N0	PIN_60	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[2]	Input	PIN_76	3	B3_N1	PIN_76	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[1]	Input	PIN_55	4	B4_N1	PIN_55	3.3-V LVTTL		24mA ...ault)	
TEC2_COL[0]	Input	PIN_58	4	B4_N1	PIN_58	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[3]	Output	PIN_51	4	B4_N1	PIN_51	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[2]	Output	PIN_52	4	B4_N1	PIN_52	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[1]	Output	PIN_47	4	B4_N1	PIN_47	3.3-V LVTTL		24mA ...ault)	
TEC2_FIL[0]	Output	PIN_48	4	B4_N1	PIN_48	3.3-V LVTTL		24mA ...ault)	
VGA_B	Output	PIN_63	4	B4_N0	PIN_63	3.3-V LVTTL		24mA ...ault)	
VGA_G	Output	PIN_65	4	B4_N0	PIN_65	3.3-V LVTTL		24mA ...ault)	
VGA_HS	Output	PIN_71	4	B4_N0	PIN_71	3.3-V LVTTL		24mA ...ault)	
VGA_R	Output	PIN_69	4	B4_N0	PIN_69	3.3-V LVTTL		24mA ...ault)	
VGA_VS	Output	PIN_72	4	B4_N0	PIN_72	3.3-V LVTTL		24mA ...ault)	
<<new node>>									

Figura 14: Asignación de pines (Pin Planner).

6. Programación de la tarjeta

La tarjeta se programó por JTAG con el USB-Blaster. Salida del programador de Quartus (sin los avisos de licencia):

```
$ lsusb | grep -i 09fb
Bus 001 Device 007: ID 09fb:6001 Altera Blaster

$ ~/altera/13.0sp1/quartus/bin/jtagconfig
1) USB-Blaster [1-4]
  020B10DD EP2C5

$ sh quartus/programar.sh
Info: Command: quartus_pgm -l
1) USB-Blaster [1-4]
Info: Quartus II 32-bit Programmer was successful. 0 errors, 0 warnings
Info: Command: quartus_pgm -m jtag -o p;output_files/sintetizador.sof
Info (213045): Using programming cable "USB-Blaster [1-4]"
Info (213011): Using programming file output_files/sintetizador.sof with checksum 0x0033EE50 for device EP2C5T144@1
Info (209060): Started Programmer operation at Tue Nov 10 12:39:24 2026
Info (209016): Configuring device index 1
Info (209017): Device 1 contains JTAG ID code 0x020B10DD
Info (209007): Configuration succeeded -- 1 device(s) configured
Info (209011): Successfully performed operation(s)
Info (209061): Ended Programmer operation at Tue Nov 10 12:39:26 2026
Info: Quartus II 32-bit Programmer was successful. 0 errors, 0 warnings
```